

# AUTOMATE

Your intelligent vehicle assistant



Final Project – Information Systems  
Engineering

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# The Problem

- Vehicle information is scattered
- Maintenance dates are easy to forget
  - Dashboard warnings can be confusing
- Vehicle documents require manual tracking
- Not every driver understands car problems

**Car ownership requires knowledge, organization and constant attention.**



# Our Solution

- One place for everything about your vehicle
  - AI-powered document analysis
  - Dashboard warning recognition
  - Smart maintenance reminders
  - Complete vehicle history

**From vehicle information → to intelligent actions**

# Target Users

- **Persona 1 : Basic Car Knowledge**

- Needs simple explanations
- Needs reminders
- Needs help understanding warnings

- **Persona 2 : Experienced Driver**

- Wants organized vehicle information
- Tracks maintenance history
- Wants quick access to vehicle data

# Core Features



## **AI Warning Scanner**

Scan dashboard warnings



## **Documents**

Upload & analyze documents



## **AI Assistant**

Ask questions about your vehicle



## **Notifications**

Smart maintenance reminders



## **Vehicle History**

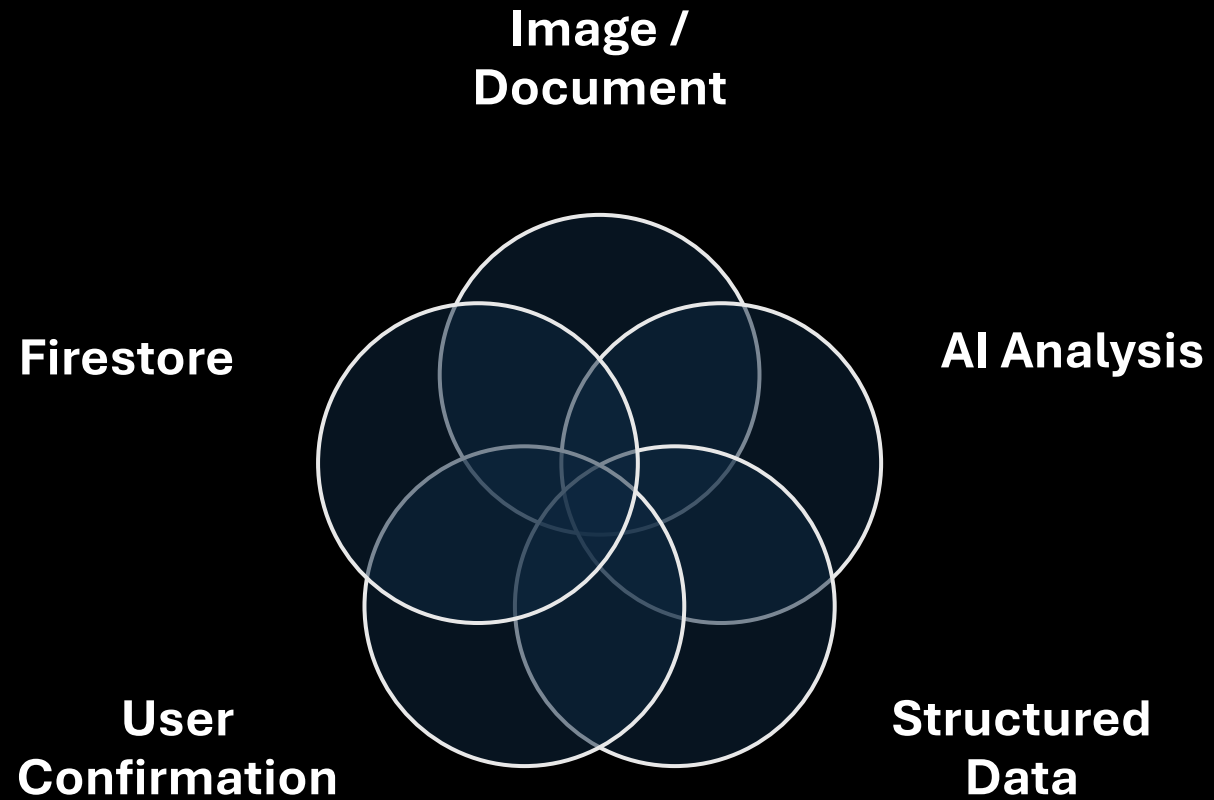
Track services & repairs



## **Vehicle Management**

Manage vehicle information

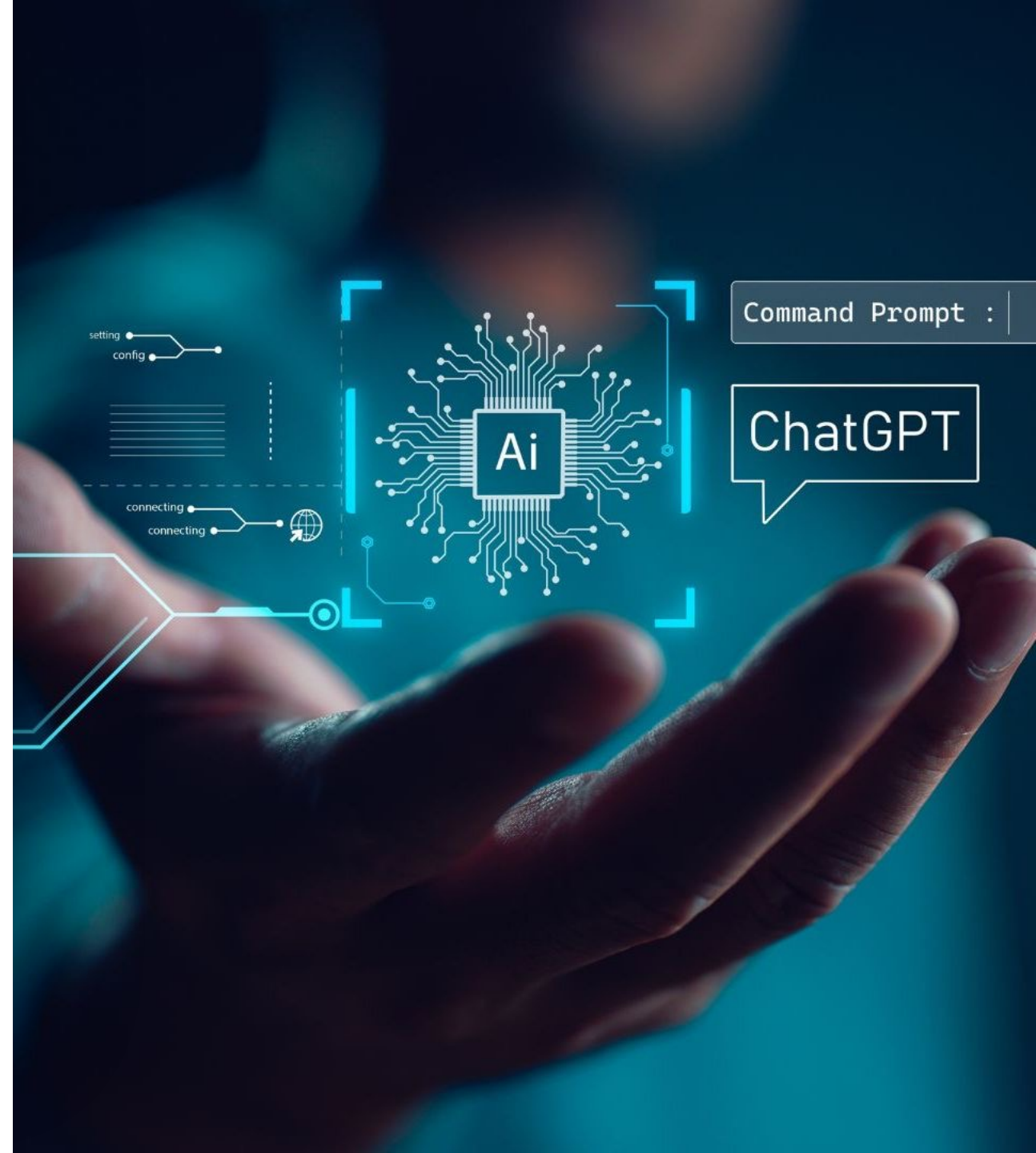
# AI-Powered Analysis



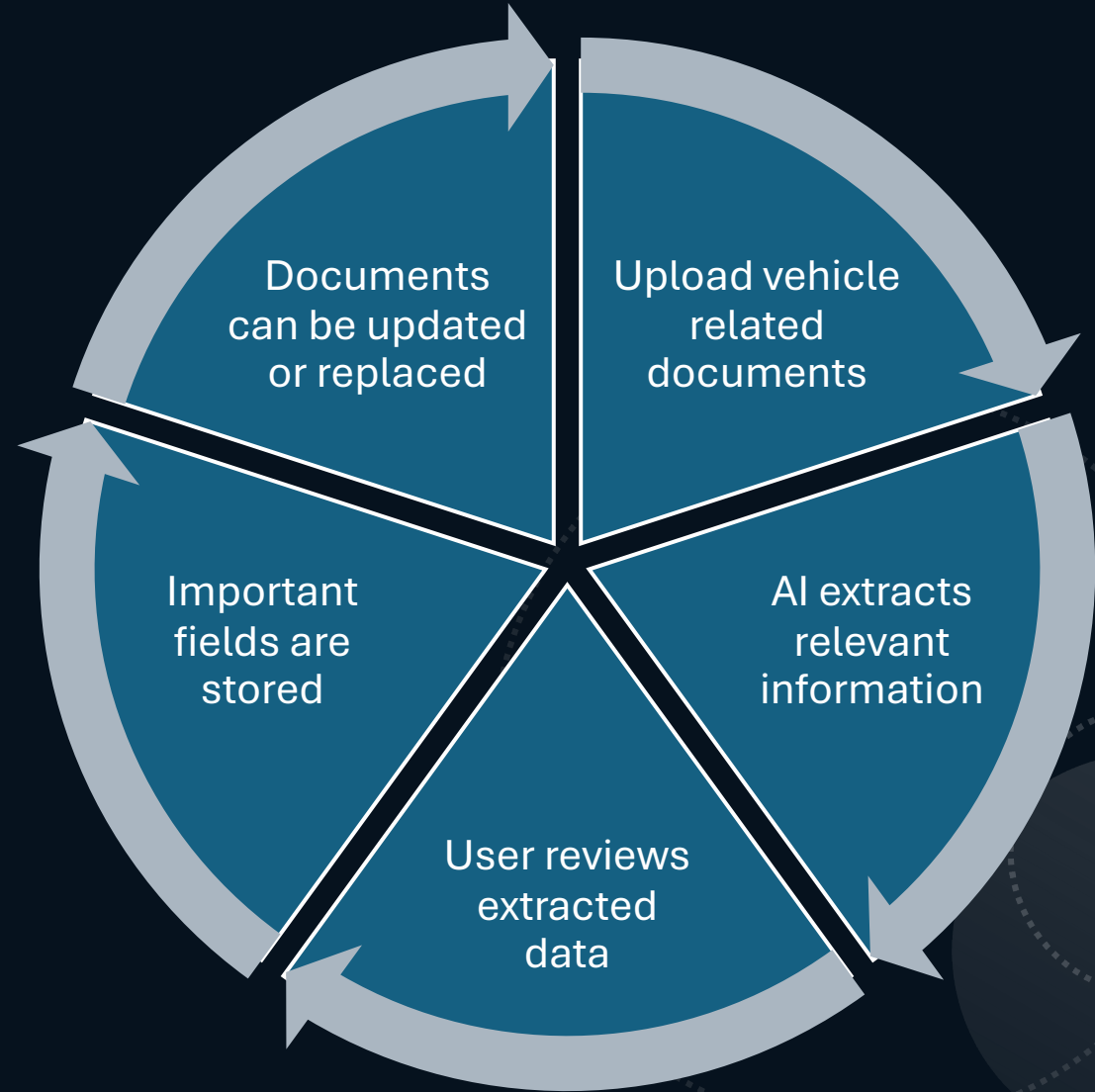
# AI Warning Scanner

- upload dashboard image
- Detect warning indicators
- Explain the warning
- Suggest recommended action
- Save confirmed event to Vehicle History

Camera → AI → Explanation → User  
Confirmation → History



# Smart Document Management



# From Data to Action

- **Vehicle History**
  - Maintenance & repairs
  - Replaced components
  - Warning events
  - Important dates
  - Complete vehicle timeline
- **Smart Notifications**
  - Vehicle inspection
  - Licence expiration
  - Maintenance
  - Upcoming service
  - Remaining time

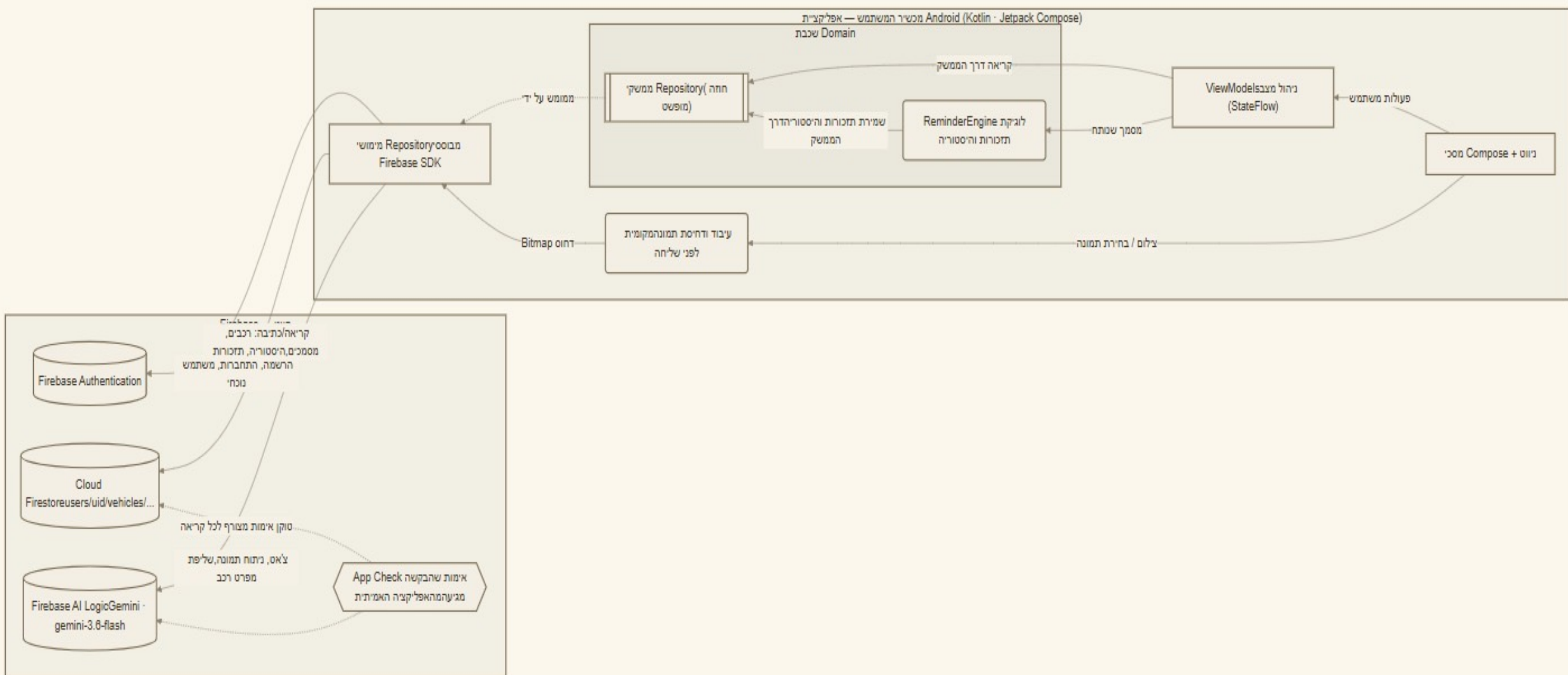
# System Architecture

## Android App

- Kotlin
- Jetpack Compose
  - ViewModel
- Repository Layer
- Firebase Services

## Technologies

- Kotlin
- Jetpack Compose
  - MVVM
- Firebase Authentication
  - Cloud Firestore
  - Firebase AI Logic
    - Gemini
- Firebase App Check



# Challenges & Lessons Learned



AI output structured application data



Firebase & Firestore synchronization



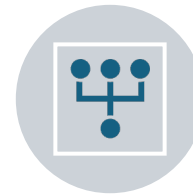
App Check / Play Integrity integration



Camera & document handling



Connecting multiple features together



The biggest challenge was not building each feature it was making them work together.

# Conclusion & Future Work

## Achievements

- Centralized vehicle management
- AI-assisted understanding
- Automated information extraction
- Connected history & reminders

## Future

- More advanced maintenance prediction
- Garage/service integration
- Improved AI accuracy
- Additional vehicle data sources